

Design of experiments

For a scientific investigation we conduct an experiment by collection of data or measurement of an object according to certain sampling procedure. Suppose we conduct an agricultural experiment to verify the truth to claim that the fertilizers increase the yield of wheat. Then the two variables, fertilizers and yield of wheat are involved directly. These two variables are called as experimental variables. In addition, quality of seed, climate, nature of soil and all other associated variables are known as extraneous variables.

The main aim of design of experiments are to control the extraneous variables,
to minimize the experimental error.

Experiment

An experiment is a device or a means of getting an answer to the problem under consideration.

Absolute experiments deals with determining the absolute value of some characteristics like obtaining the average intelligence quotient of a group of people.

Comparative experiments are designed to compare the effect of two or more objects on some population characteristics.

eg. Comparison of different kinds of varieties of crops.

Treatment.

Various objects of comparison in a comparative experiment are termed as treatment.

eg. In a field experiment, different fertilizers, dif. varieties of crop, diff. methods of cultivation.

Experimental unit.

The smallest division of the experimental material to which we apply the treatments and on which we make observations of the variable under study is termed as experimental unit.

eg. land

Yield

The measurement of the variable under study on different experimental units are termed as yields.

Experimental error.

- It includes all types of extraneous variations due to
- i. the inherent variability in the experimental material to which treatments are applied.
 - ii. the lack of uniformity in the methodology of conducting the experiment or in other words failure of standardised experimental techniques.

To control the effect of extraneous variables, we use grouping and blocking.

By grouping we mean combining sets of homogeneous experimental units. The different groups need not have the same no. of "

By blocking we mean assigning same no. of experimental units in different blocks. Each block will have comparatively homogeneous experimental units.

Completely Randomised Design,

Randomised Block " ,

Latin square
 2^k Factorial design, Taguchi's robust parameter design
CRD.

CRD is the simplest of all the designs based on the principles of randomisation and replication. In this design, treatments are allocated at random to experimental units over the entire experimental material.

Let us suppose that we have v treatments. The i th treatment being replicated r_i times ($i=1$ to v). Then the whole experimental material is divided into $n = \sum r_i$ experimental units and the treatments are distributed completely at random over the units subject to the condition that the i th treatment occurs r_i times. Randomization assures that extraneous factors do not continuously influence one factor.

Advantages of CRD.

There is a complete flexibility in the model as the no. of replications is not fixed.

Analysis can be performed even if some observations are missing.

CRD results in the maximum use of the experimental units since all the experimental material can be used.

Disadvantage.

The experimental error is large as compared to the other designs since the homogeneity of the unit is ignored.

Statistical analysis of CRD.

ANOVA is a technique used to test the means of more than two samples. It divides the total variance in the group into parts, which are associated to different factors. This variation is split into two components as
variation within subgroups.
variation between the subgroups.

Model eqn.

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad 1 \leq i \leq v, \quad 1 \leq j \leq n \text{ where}$$

y_{ij} - yield or response from the j th unit receiving the i th treatment

α_i - effect due to the i th treatment.

ϵ_{ij} - error effect.
(independent normally distributed random variables with zero means and common variance $= \sigma^2$)

H_0 : The v pop means are equal
or $\alpha_1 = \alpha_2 = \dots = \alpha_v = 0$.

H_1 : At least two of the pop means are unequal.
 $\alpha_i \neq 0$ for some i .

ANOVA Table

Source of variation	Degrees of freedom	Sum of squares	Mean Square	F-ratio
Treatments	$v-1$	SST	$MST = \frac{SST}{v-1}$	$F = \frac{MST}{MSE}$
Error	$n-v$	SSE	$MSE = \frac{SSE}{n-v}$	
Total	$n-1$	$SST + SSE$		

where v - no. of treatments

$n = \sum_{i=1}^v r_i$ - total no. of experimental units

$$SST = \sum_{i=1}^v \frac{T_i^2}{r_i} - C.F. \quad , \quad C.F. - \text{Correction Factor} = \frac{G^2}{n}, \quad G = \text{Grand total}$$

$$SSE = TSS - SST,$$

For convenience we denote

$$SST = Q_1, \quad SSE = Q_2$$

$$TSS = \sum_j \sum_i y_{ij}^2 - C.F.$$

$$\text{and } TSS = Q_1 + Q_2$$

Table value $F_{\alpha}(v-1, n-v)$

1. A set of data involving four tropical feeds A, B, C, D tried on 20 chicks are given below. All the 20 chicks are treated alike in all aspects except the feeding treatment. Each feeding treatment is given to 5 chicks. Analyse the data.

	Weight gain					Total
A	55	49	42	21	52	219
B	61	112	30	89	63	355
C	42	97	81	95	92	407
D	169	137	169	85	154	714
						<u>1695</u>

H_0 : Effects due to four feeds are equal.

$$i. H_0: \mu_A = \mu_B = \mu_C = \mu_D$$

$$H_1: \mu_A \neq \mu_B \neq \mu_C \neq \mu_D$$

$$G = 1695, \quad n = 20, \quad v = 4, \quad r_i = 5, \quad 1 \leq i \leq 4$$

$$C.F. = \frac{G^2}{n} = 143651.25$$

$$TSS = 181445 - 143651.25 = 37793.75$$

$$SST = 169886.2 - \quad \quad \quad = 26234.95$$

$$SSE = 11558.8$$

ANOVA Table

Source of variation	Degrees of freedom	Sum of squares	Mean square	F
Treatment	$V-1 = 3$	26234.95	8744.983	12.105
Error	$n-V = 16$	11558.8	722.425	
Total	19	37793.95		

$$F_{0.05}(3, 16) = 3.24$$

$$F > F_{\alpha}$$

$\therefore$ Reject H_0 .

2. Set the ANOVA table for the following per hectare yield for three varieties of wheat each grown in 4 plots.

Plots	Varieties of wheat			
	A_1	A_2	A_3	
1	6	5	5	
2	7	5	4	
3	3	3	3	
4	8	7	4	
Total	24	20	16	60

Test whether there is a significant dif. among yield in ^{the} three varieties of wheat.

$$H_0: \mu_{A_1} = \mu_{A_2} = \mu_{A_3}$$

$$G = 60, V = 3, n = 12.$$

$$C.F = 360$$

$$TSS = 332 - 360 = 32$$

$$SST = 8$$

$$SSE = 24$$

Source of variation	Degrees of freedom	S.S	M.S	F
Treatment	2	8	4	1.5003
Error	9	24	2.666	
Total	11	32		

$$F_{0.05}(2,9) = 4.26$$

$$F < F_{\alpha}$$

$\therefore$ Accept H_0 .

3. A completely randomised design experiment with 10 plots and 3 treatments gave the following results:

Plot No.	1	2	3	4	5	6	7	8	9	10
Treatment	A	B	C	A	C	C	A	B	A	B
Yield	5	4	3	7	5	1	3	4	1	7

Analyse the results for treatment effects.

Solution:

Rearranging the data according to the treatments, we have the following table:

Treatment	Yield from plots (x_{ij})				T_i	T_i^2	n_i	$\frac{T_i^2}{n_i}$
A	5	7	3	1	16	256	4	64
B	4	4	7	—	15	225	3	75
C	3	5	1	—	9	81	3	27
Total					$T = 40$	—	$N = 10$	166

$$\begin{aligned}\sum \sum x_{ij}^2 &= (25 + 49 + 9 + 1) + (16 + 16 + 49) + (9 + 25 + 1) \\ &= 84 + 81 + 35 = 200\end{aligned}$$

$$Q = \sum \sum x_{ij}^2 - \frac{T^2}{N} = 200 - \frac{40^2}{10} = 200 - 160 = 40$$

$$Q_1 = \sum \frac{T_i^2}{n_i} - \frac{T^2}{N} = 166 - 160 = 6$$

$$Q_2 = Q - Q_1 = 40 - 6 = 34$$

ANOVA table

S.V.	S.S.	d.f.	M.S.	F_0
Between classes (treatments)	$Q_1 = 6$	$h - 1 = 2$	3.0	$\frac{4.86}{3.0}$
Within classes	$Q_2 = 34$	$N - h = 7$	4.86	$= 1.62$
Total	$Q = 40$	$N - 1 = 9$	—	—

From the F -table, $F_{5\%}(v_1 = 2, v_2 = 7) = 19.35$

We note that $F_0 < F_{5\%}$

Let H_0 : The treatments do not differ significantly.

$\therefore$ The null hypothesis is accepted.

i.e., the treatments are not significantly different.

4. A manufacturing company has purchased 3 new machines of different makes and wish to determine whether one of them is faster than the others in producing certain output. Five hourly production figures are observed at random from each machine and the results are given below.

Observations	A_1	A_2	A_3
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	25
5	31	35	28

we analysis of variance and determine whether the machines are different in their mean speed.

Ans: Here $N = 15$, $h = 3$, $n_1 = n_2 = n_3 = 5$

H_0 : The machines are don't differ significantly.

H_1 : The machines are differ significantly.

Table:

machines	observations	T_i	T_i^2	n_i	$\frac{T_i^2}{n_i}$
A_1	25 30 36 38 31	160	25600	5	5120
A_2	31 39 38 42 35	185	34225	5	6845
A_3	24 30 28 25 28	135	18225	5	3645
Total		$T = 480$	-	$N = 15$	15610

$\Rightarrow \sum \sum x_{ij}^2 = 15810$

$$Q = \sum \sum x_{ij}^2 - \frac{T^2}{N} = 15810 - \frac{480^2}{15} = 450$$

$$Q_1 = \sum_i \frac{T_i^2}{n_i} - \frac{T^2}{N} = \frac{160^2}{5} + \frac{185^2}{5} + \frac{135^2}{5} - \frac{480^2}{15} = 5120 + 6845 + 3645 - 15360 = 15610 - 15360 = 250$$

$$Q_1 = 250$$

$$Q_2 = Q - Q_1 = 450 - 250 = 200$$

$$h = 3$$

∴ Table: (ANOVA)

S.S.	S.S	d.f	m.s.s	F
between machine	$Q_1 = 250$	$h-1 = 2$	$\frac{250}{2} = 125$	$F = \frac{125}{16.67} = 7.50$
within machine	$Q_2 = 200$	$N-h = 15-3 = 12$	$\frac{200}{12} = 16.67$	
Total	450	14		

$$\Rightarrow F_{cal} = 7.50$$

$$\text{from table: } F_{5\%}(v_1=2, v_2=12) = 3.89$$

Since $F_{cal} > F_{tab}$

∴ we reject H_0 .

That is There is a significance difference in the three machine.

Exercise:

The following table shows the lives in hours of four brands of electric lamps:

Brand

A : 1610, 1610, 1650, 1680, 1700, 1720, 1800

B : 1580, 1640, 1640, 1700, 1750

C : 1460, 1550, 1600, 1620, 1640, 1660, 1740, 1820

D : 1510, 1520, 1530, 1570, 1600, 1680

Perform an analysis of variance and test the homogeneity of the mean lives of the four brands of lamps.

Randomised Block Design (2 way classification)

If the whole experimental area is not homogenous, then a simple method of controlling the variability of experimental material consists in grouping the whole area into relatively homogenous subgroups.

The treatments can be applied in a random manner to relatively homogenous units within each subgroup/block and replicated over all the blocks. This design is known as RBD.

Layout.

Let us consider 5 treatments A, B, C, D, E each replicated four times. We divide the whole experimental area into 4 relatively homogenous blocks and each block into 5 units. Treatments are allocated at random to the plots of blocks.

Blocks

<u>I</u>	A	E	B	D	C
<u>II</u>	E	D	C	B	A
<u>III</u>	C	B	A	E	D
<u>IV</u>	A	D	E	C	B

Advantages.

1. It has a simple layout
 2. This design controls the variability in the experimental units and gives the treatments equivalence to show their effects.
- Disadv. It is not suitable for large no. of treatments.

Model eqn.

$$y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \quad , \quad 1 \leq i \leq a \\ 1 \leq j \leq b$$

a - no. of treatments

b - " " or blocks

μ - grand mean

α_i - effect of i^{th} treatment

β_j - " " j^{th} block

ε_{ij} - independent normally distributed random variable having mean 0 and variance σ^2 .

ANOVA Table

Source of variation	Degrees of freedom	Sum of Squares	M.S.	F
Treatments	$a-1$	SST	$MST = \frac{SST}{a-1}$	$F_T = \frac{MST}{MSE}$
Blocks	$b-1$	SSB	$MSB = \frac{SSB}{b-1}$	$F_B = \frac{MSB}{MSE}$
Error (residual)	$(a-1)(b-1)$	SSE	$MSE = \frac{SSE}{(a-1)(b-1)}$	
Total	$ab-1$	TSS		

$$C.F = \frac{G_1^2}{ab}$$

$$SST = \sum_{i=1}^a T_i^2 - C.F$$

$$SSB = \sum_{j=1}^b B_j^2 - C.F$$

$$TSS = \sum_j \sum_i y_{ij}^2 - C.F$$

$$SSE = TSS - (SST + SSB)$$

If $F_T < F_{\alpha}((a-1), (a-1)(b-1))$, accept H_0

If $F_B < F_{\alpha}((b-1), (a-1)(b-1))$, accept H_0

For convenience we denote

$$SST = Q_1, \quad SSB = Q_2, \quad SSE = Q_3 \quad \text{and} \quad TSS = Q_1 + Q_2 + Q_3$$

1. Consider the results given in the fol. table involving 6 treatments in 4 randomised blocks. The treatments indicated by numbers within parenthesis. Analyse whether there is any significant difference between the treatments and blocks are homogenous.

Blocks	Treatments and yield					
	(1)	(2)	(3)	(4)	(5)	(6)
1	24.7	27.7	20.6	16.2	16.2	24.9
2	22.9	28.8	27.3	15	22.5	17
3	26.3	19.6	38.5	36.8	39.5	15.4
4	17.7	31	28.5	14.1	34.9	22.6

$$H_T : \alpha_1 = \alpha_2 = \dots = \alpha_6$$

$$H_B : \beta_1 = \beta_2 = \dots = \beta_4$$

	(1)	(2)	(3)	(4)	(5)	(6)	Total
1	24.7	27.7	20.6	16.2	16.2	24.9	130.3
2	22.9	28.8	27.3	15	22.5	17	133.5
3	26.3	19.6	38.5	36.8	39.5	15.4	176.1
4	17.7	31	28.5	14.1	34.9	22.6	148.8
Total	119	127	115.2	64.9	66.3	96.3	588.7

$$C.F = 14,440.32$$

$$TSS = 15789.89 - C.F = 1349.57$$

$$SST = 15360.6075 - C.F = 920.28$$

$$SSB = 14658.831 - " = 218.51$$

$$SSE = 210.78$$

Treatment	5	920.28	184.056	13.098
Block	3	218.51	72.836	5.1833
Error	15	210.78	14.052	

$$F_{\alpha}(5, 15) = 2.9 \quad F_T > F_{\alpha}(5, 15)$$

$$F_{\alpha}(3, 15) = 3.29 \quad F_B > F_{\alpha}(3, 15)$$

Reject H_T, H_B

2. An experiment was designed to study the performance of 4 different detergents for cleaning fuel injectors. The following cleanliness readings were obtained with specially designed equipment for 12 tanks of gas distributed over 3 different models of engines.

Detergent	Engine		
	1	2	3
A	45	43	51
B	47	46	52
C	48	50	55
D	42	37	49

Looking on the detergents as treatments and the engines as blocks, obtain the two way analysis of variance table and test at 1% level of significance whether there are differences in the detergents or in the engine.

3. Three varieties of a crop are tested in a randomised block design with four replications, the layout being as given below: The yields are given in kilograms. Analyse for significance

C48	A51	B52	A49
A47	B49	C52	C51
B49	C53	A49	B50

Solution:

Rewriting the data such that the rows represent the blocks and the columns represent the varieties of the crop (as assumed in the discussion of analysis of variance for two factors of classification), we have the following table:

Crops

<i>Blocks</i>	<i>A</i>	<i>B</i>	<i>C</i>
1	47	49	48
2	51	49	53
3	49	52	52
4	49	50	51

We shift the origin to 50 and work out with the new values of x_{ij} .

Crops

<i>Blocks</i>	<i>A</i>	<i>B</i>	<i>C</i>	T_i	T_i^2/k	$\sum x_{ij}^2$
1	-3	-1	-2	-6	36/3 = 12	14
2	1	-1	3	3	9/3 = 3	11
3	-1	2	2	3	9/3 = 3	9
4	-1	0	1	0	0/3 = 0	2
T_j	-4	0	4	$T = 0$	$\sum \frac{T_i^2}{k} = 18$	36
T_j^2/h	$\frac{16}{4} = 4$	$\frac{0}{4} = 0$	$\frac{16}{4} = 4$	$\sum \frac{T_j^2}{h} = 8$		
$\sum x_{ij}^2$	12	6	18	36		

$$Q = \sum \sum x_{ij}^2 - \frac{T^2}{N} = 36 - \frac{0^2}{12} = 36$$

$$Q_1 = \frac{1}{k} \sum T_i^2 - \frac{T^2}{N} = 18 - 0 = 18$$

$$Q_2 = \frac{1}{h} \sum T_j^2 - \frac{T^2}{N} = 8 - 0 = 8$$

$$Q_3 = Q - Q_1 - Q_2 = 36 - 18 - 8 = 10$$

ANOVA table

S.V.	S.S.	d.f.	M.S.	F_0
Between rows (blocks)	$Q_1 = 18$	$h - 1 = 3$	6	$\frac{6}{1.67} = 3.6$
Between columns (crops)	$Q_2 = 8$	$k - 1 = 2$	4	$\frac{4}{1.67} = 2.4$
Residual	$Q_3 = 10$	$(h - 1)(k - 1) = 6$	1.67	-
Total	$Q = 36$	$hk - 1 = 11$	-	-

From F -tables, $F_{5\%}(v_1 = 3, v_2 = 6) = 4.76$ and $F_{5\%}(v_1 = 2, v_2 = 6) = 5.14$. Considering the difference between rows, we see that $F_0 (= 3.6) < F_{5\%} (= 4.76)$. Hence the difference between the rows is not significant. (H_0 is accepted) viz., the blocks do not differ significantly with respect to the yield.

Considering the difference between columns, we see that $F_0 (= 2.4) < F_{5\%} (= 5.14)$.

Hence the difference between the columns is not significant. (H_0 is accepted) viz., the varieties of crop do not differ significantly with respect to the yield.

4. A tea company appoints four salesmen A, B, C and D and observes their sales in three seasons - summer, winter and monsoon. The figures (in lakhs) are given in the following table.

Season	Salesman				Total
	A	B	C	D	
Summer	36	36	21	25	128
Winter	28	29	31	32	120
monsoon	26	28	29	29	112
Total	90	93	81	96	360

(a) Do the salesmen significantly differ in performance?

(b) Is there a significant difference between the salesmen

Ans:

$$N=12$$

We shift the origin to 30 by subtracting to each.

Seasons	Salesman				T_i	T_i^2/k	$\sum x_{ij}^2$
	A	B	C	D			
Summer	6	6	-9	5	8	$\frac{64}{4}=16$	178
Winter	-2	-1	1	2	0	$\frac{0}{4}=0$	10
monsoon	-4	-2	-1	-1	-8	$\frac{64}{4}=16$	22
T_j	0	3	-9	6	$T=0$	$\frac{1}{k} \sum T_i^2 = 32$	210
T_j^2/h	$\frac{0}{3}=0$	3	27	12	$\frac{\sum T_j^2}{h}=42$		
$\sum x_{ij}^2$	56	41	83	30			$\sum \sum x_{ij}^2 = 210$

$$\therefore Q = \sum \sum x_{ij}^2 - \frac{T^2}{N} = 210 - \frac{0^2}{12} = 210$$

$$Q_1 = \frac{1}{k} \sum T_i^2 - \frac{T^2}{N} = 32 - \frac{0^2}{12} = 32$$

$$Q_2 = \frac{1}{h} \sum T_j^2 - \frac{T^2}{N} = 42 - 0 = 42$$

$$Q_3 = Q - Q_1 - Q_2 = 210 - 32 - 42 = 136$$

S.V.	S.S.	d.f	m.s.	F
Between rows (Seasons)	$Q_1 = 32$	$h-1 = 2$	16	$F = 1.4125$
Between cols (Salesman)	$Q_2 = 42$	$k-1 = 3$	14	$F = 1.6142$
Residual.	$Q_3 = 136$	$(h-1)(k-1) = 6$	$\frac{136}{6} = 22.6$	
Total	210	$hk-1 = 11$		

Consider the seasons:

$$F_{cal} = 1.4125 \Rightarrow F_{cal} < F_{tab}$$

$$F_{tab} = F_{5,2}(6,2) = 19.33$$

$\Rightarrow H_0$ accepted.

There is no difference in performance

Consider the Salesmen:

$$F_{cal} = 1.6142$$

$$\Rightarrow F_{cal} < F_{tab}$$

$$F_{tab} = F_{5,3}(6,3) = 8.94$$

$\Rightarrow H_0$ is accepted

There is no significant difference
in Salesmen.

Exercise:

The following data represent the number of units of production per day turned out by 5 different workers using 4 different types of machines:

		Machine Type			
		A	B	C	D
Workers:	1	44	38	47	36
	2	46	40	52	43
	3	34	36	44	32
	4	43	38	46	33
	5	38	42	49	39

- Test whether the five men differ with respect to mean productivity.
- Test whether the mean productivity is the same for the four different machine types.

Latin Square design

It is used to eliminate the effects of 2 extraneous sources of variability. An $n \times n$ Latin square is a square array of n distinct letters, with each letter appearing once and only once in each row and each column.

Layout.

Suppose there are four treatments A, B, C, D each applied once in each row and each column.

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

4 x 4 Latin Square

Advantages.

1. Latin square design controls more of the variation than RBD.
2. Analysis is simple.
3. Even with missing data the analysis remains relatively simple.

Disadv. It cannot be applied for all experiments

Model equation.

$$y_{ijk} = \mu + \alpha_i + \beta_j + \tau_k + \epsilon_{ijk}, \quad i, j, k = 1, 2, \dots, m$$

y_{ijk} - yield obtained from the i th row and j th column

by applying k th treatment.

μ - grand mean

α_i - effect of the i th row

β_j - " " " j th column

τ_k - " " " k th treatment

ϵ_{ijk} - indep. normally distributed random variables with zero means and common variance σ^2 .

$$H_R: \alpha_i = 0 \quad \forall i \quad H_{R_1}: \text{Atleast one } \alpha_i \neq 0$$

$$H_C: \beta_j = 0 \quad \forall j$$

$$H_T: \gamma_k = 0 \quad \forall k$$

ANOVA Table

Source of variation	Degrees of freedom	S.S	MS	F
Row	$m-1$	SSR	$MSR = \frac{SSR}{m-1}$	$F_R = MSR/MSE$
Column	$m-1$	SSC	$MSC = \frac{SSC}{m-1}$	$F_C = MSC/MSE$
Treatment	$m-1$	SST	$MST = \frac{SST}{m-1}$	$F_T = MST/MSE$
Error	$(m-1)(m-2)$	SSE	$MSE = \frac{SSE}{(m-1)(m-2)}$	
Total	m^2-1	TSS		

$$C.F = \frac{G^2}{m^2}, \quad m = \text{No. of rows} = \text{No. of columns} = \text{No. of treatments}$$

$$SSR = \frac{\sum R_i^2}{m} - C.F$$

$$SSC = \frac{\sum C_j^2}{m} - C.F$$

$$SST = \frac{\sum T_k^2}{m} - C.F$$

$$TSS = \sum \sum y_{ij}^2 - C.F$$

$$SSE = TSS - (SSR + SSC + SST)$$

$$\text{If } F_R < F_{\alpha}(m-1, (m-1)(m-2)), \text{ accept } H_R$$

$$\text{" } F_C \text{ " " " " } H_C$$

$$\text{If } F_T \text{ " " " " } H_T$$

1. Set up the ANOVA for the following results of a Latin square design

A	C	B	D	49
12	19	10	8	
C	B	D	A	43
18	12	6	7	
B	D	A	C	58
22	10	5	21	
D	A	C	B	63
12	7	27	17	
64	48	48	53	213

SV	Df	SS	MSS	F
R	3	60.1875	20.0625	1.5165
C	3	42.6875	14.2292	1.0756
T	3	465.1875	155.0625	11.7212
E	6	79.375	13.2292	

$$F_{0.05}(3, 6) = 4.76$$

Accept H_R , H_C , Reject H_T .

Five doctors each test five treatments for a certain disease and observe the no. of days each patient takes to recover. Discuss the diff between
 i) the doctors ii) the treatments for the fol. data

Doc.	1	2	3	4	5	Treat.	4	406.64	101.66	470.6
1	10	14	23	18	20	Doc.	4	2584	6.46	2.99
2	11	15	24	17	21	Error	16	34.56	2.16	
3	9	12	20	16	19			467.04		
4	8	13	17	17	20					
5	12	15	19	15	22					

$$F_{\alpha}(4, 16) = 3.01$$

2. The sample data in the following Latin Square are the scores obtained by 9 college students of various ethnic backgrounds and various professional interests in an American history test. A, B, C are the three instructors by whom the 9 college students were taught the course in American history. Use 5% level of significance to analyze the design and test the following hypotheses. whether differences in
- the ethnic background have no effect on the scores and
 - professional interests have no effect on the scores.

	Ethnic background		
	Mexican	German	Polish
Law	A: 75	B: 86	C: 69
Medicine	B: 95	C: 79	A: 86
Engineering	C: 70	A: 83	B: 93

		SS	MSS	F	
SV	Df		75.11	27.11	2.407
Row	2	150.23			1.472
Column	2	14.23	7.11	2.567	2.227
Treatment	2	523.56	261.78	94.505	
Error	2	5.54	2.77		
Total	8	693.56			

$$F_{0.05}(2, 2) = 19$$

Accept H_0

Reject H_R, H_T